



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

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Div: CSE A1

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ASSIGNMENT No : 8

TITLE :Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance. **Learning -**

Objectives

To demonstrate the use of the super keyword for accessing parent class members. **Theory**

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Outcome

The program demonstrated the use of the super keyword to invoke the parent class constructor and access parent class members from within a subclass. The output confirmed correct constructor chaining and successful retrieval of inherited data.

Conclusion

The super keyword is essential in inheritance for reusing parent class initialization logic and resolving naming conflicts between parent and child members. This experiment showed how it strengthens the relationship between base and derived classes.

Exercise

1. Create an **Employee and Manager program** where Manager accesses Employee details using super keyword.

```
1  class Employee {
2      String name;
3      double salary;
4      Employee(String name, double salary) {
5          this.name = name;
6          this.salary = salary;
7      }
8      void display() {
9          System.out.println("Employee: " + name + ", Salary: " + salary);
10     }
11 }
12 class Manager extends Employee {
13     double bonus;
14
15     Manager(String name, double salary, double bonus) {
16         super(name, salary);
17         this.bonus = bonus;
18     }
19
20     void display() {
21         super.display();
22         System.out.println("Bonus: " + bonus);
23     }
24 }
25 public class EmployeeDemo {
26     Run | Debug
27     public static void main(String[] args) {
28         Manager m = new Manager(name: "Rahul", salary: 50000, bonus: 5000);
29         m.display();
30     }
31 }
```

Employee: Rahul, Salary: 50000.0
Bonus: 5000.0

2. Develop a **Vehicle Insurance System** where child insurance classes access parent vehicle information using the super keyword.

```
1  class Vehicle {
2      String model;
3      Vehicle(String model) { this.model = model; }
4      void showVehicle() { System.out.println("Vehicle Model: " + model); }
5  }
6  class Insurance extends Vehicle {
7      double premium;
8      Insurance(String model, double premium) {
9          super(model);
10         this.premium = premium;
11     }
12     void showInsurance() {
13         super.showVehicle();
14         System.out.println("Insurance Premium: " + premium);
15     }
16 }
17 public class InsuranceDemo {
18     Run | Debug
19     public static void main(String[] args) {
20         Insurance ins = new Insurance(model: "Honda City", premium: 12000);
21         ins.showInsurance();
22     }
23 }
```

```
Vehicle Model: Honda City
Insurance Premium: 12000.0
```
